# Trigonometry in 3D

**1** All edges of the following cube are $7$ cm long:

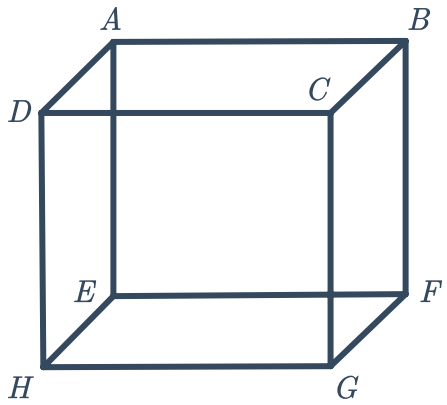
**a** Find the exact length of the following:

i $EG$ ii $AG$

**b** Find the following angles to the nearest degree:

i $\angle EGH$ ii $\angle EGA$

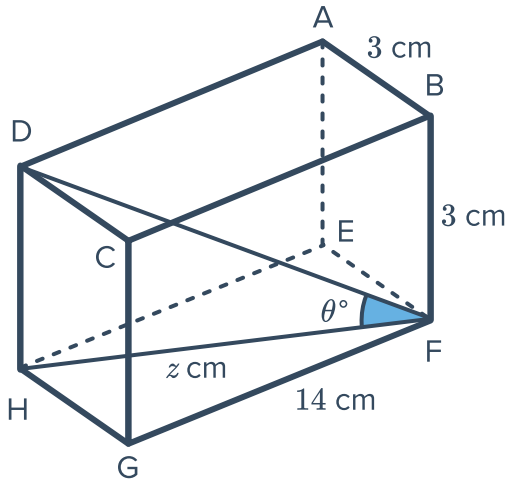
iii $\angle AHG$ iv $\angle AGH$




**2** A square prism has sides of length $3$ cm, $3$ cm and $14$ cm as shown in the diagram:

**a** If the diagonal $HF$ has a length of $z$ cm, find the value of $z$ to two decimal places.

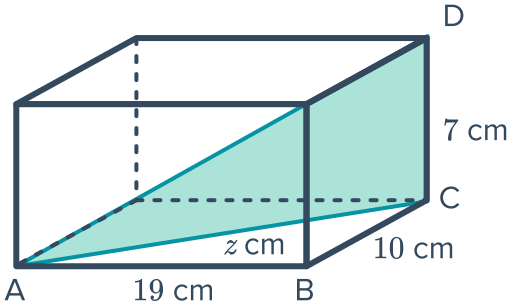
**b** If the size of $\angle DFH$ is $\theta^\circ$, find $\theta$ to two decimal places.




**3** The box have a triangular divider placed inside it, as shown in the diagram:

**a** If $z = AC$, find the value of $z$ to two decimal places.

**b** Find the area of the divider correct to two decimal places.




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**4** A triangular prism has dimensions as shown in the diagram:

**a** Find the size of $\angle AED$, correct to two decimal places.

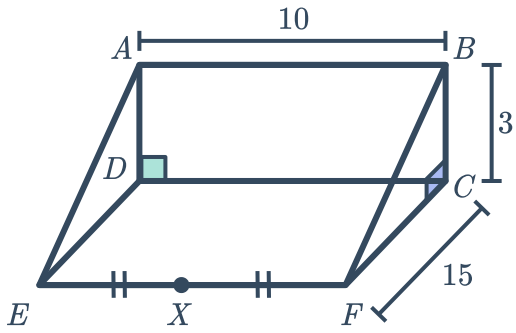
**b** Find the exact length of $CE$.

**c** Find the exact length of $BE$.

**d** Find the size of $\angle BEC$, correct to two decimal places.

**e** Find the exact length of $CX$.

**f** Find the size of $\angle BXC$, correct to two decimal places.

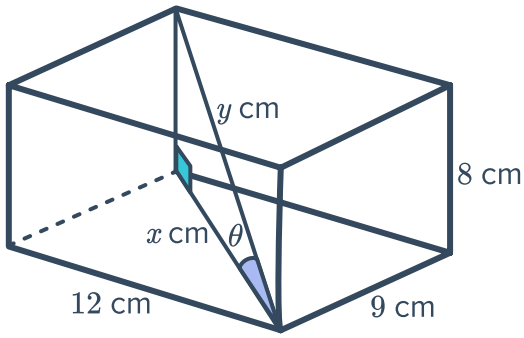



**5** Consider the given rectangular prism:

**a** Find the length $x$.

**b** Hence, find the length of the prism's diagonal $y$.

**c** Find the angle $\theta$ to the nearest degree.




**6** Consider the cube as shown:

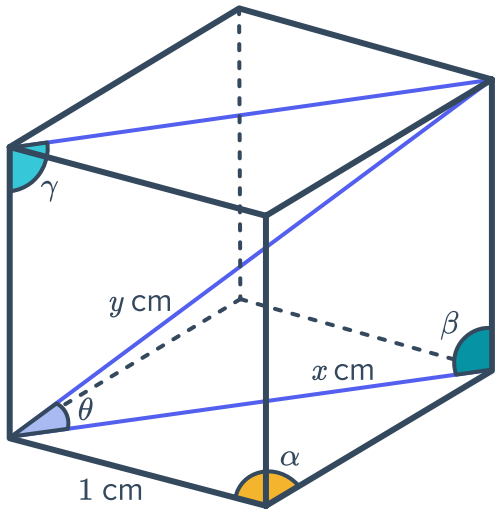
**a** Find the size of:

i $\alpha$ ii $\beta$ iii $\gamma$

**b** Calculate the exact length of $x$.

**c** Calculate the exact length of $y$.

**d** Hence, find the angle $\theta$ to the nearest degree.



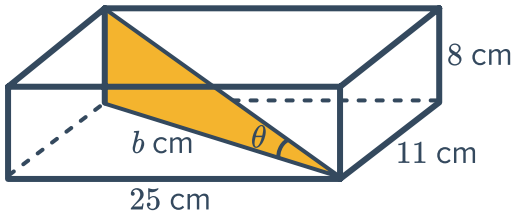

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**7** A $25\text{ cm} \times 11\text{ cm} \times 8\text{ cm}$ cardboard box contains an insert (the shaded area) made of foam:

**a** Find the exact length $b$ of the base of the foam insert.

**b** Find the area of foam in the insert, rounded to the nearest square centimetre.

**c** Find the value of $\theta$ to the nearest degree.



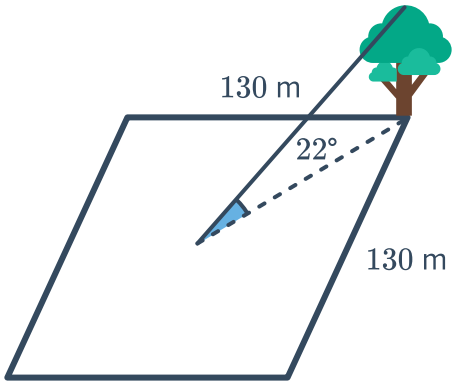

**8** A tree stands at the corner of a square playing field. Each side of the square is $130\text{ m}$ long. At the centre of the field the tree subtends an angle of $22^\circ$ as shown:

**a** Find the distance from the tree to the centre of the field. Round your answer to two decimal places.

**b** Find the height of the tree. Round your answer to two decimal places.

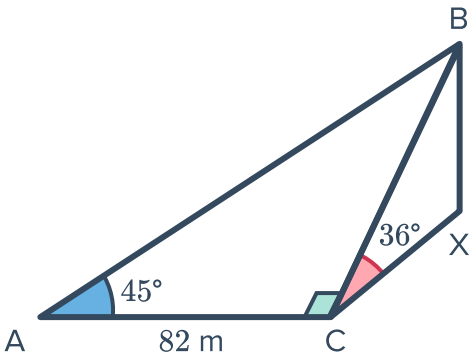
**c** Find the size of the angle subtended by the tree at the corner of the field opposite the tree. Round your answer to the nearest minute.

**d** Find the size of the angle subtended by the tree at an adjacent corner of the field. Round your answer to the nearest minute.



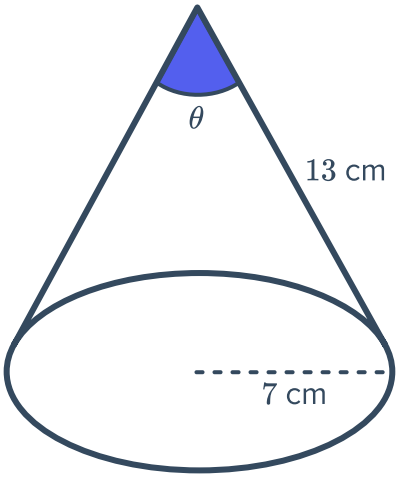

**9** $A$, $C$, and $X$ are three points in a horizontal plane and $B$ is a point vertically above $X$ as shown in the diagram:
Find the following lengths to the nearest metre:

**a** $BC$ **b** $XB$




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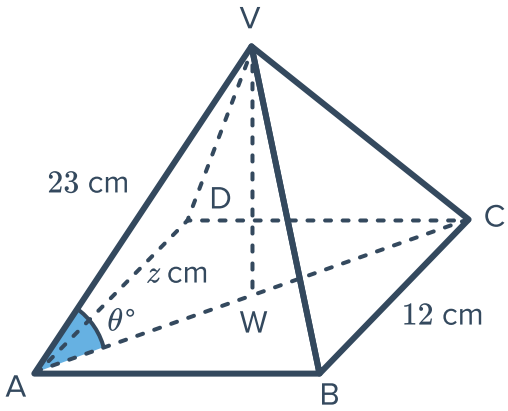
**10** A cone has radius $7$ cm and a slant height of $13$ cm:
Find the vertical angle, $\theta$, at the top of the cone in degrees and minutes.




**11** The following is a right pyramid on a square base with side length $12$ cm. The edge length $VA$ is $23$ cm.

**a** If $z = AW$, calculate the length $z$.

**b** If $\theta = \angle VAW$, find the value of $\theta$.




**12** The following pyramid has a square base with side length $8$ cm, and all slanted edges are $12$ cm in length:

**a** Find the exact length of $MD$.

**b** Find the size of $\angle PDM$, correct to two decimal places.

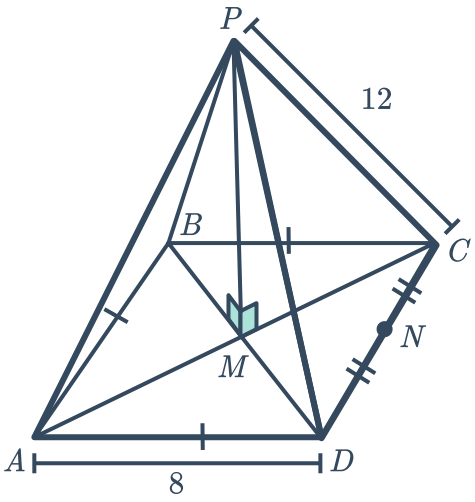
**c** Find the exact height of the pyramid, $MP$.

**d** Find the length of $MN$.

**e** Find the exact length of $PN$.

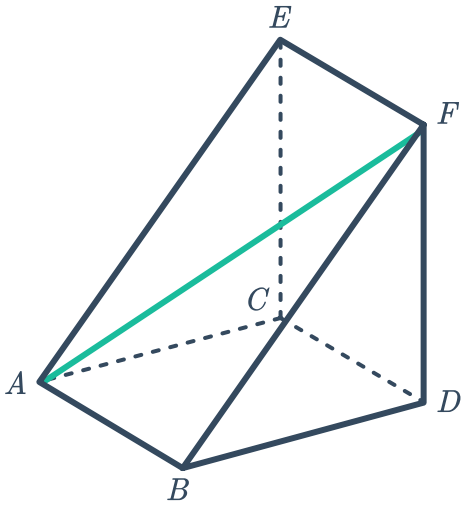
**f** Find the size of $\angle PNM$, correct to two decimal places.

**g** Find the size of $\angle PDC$, correct to two decimal places.




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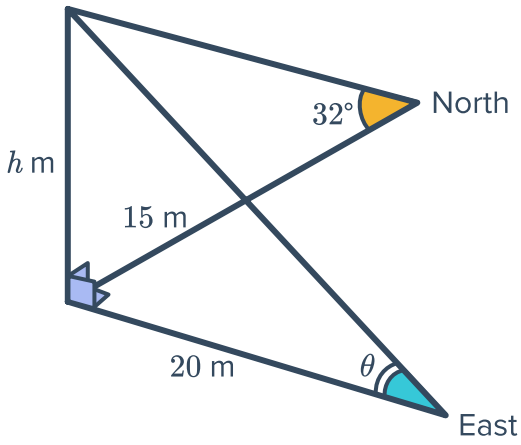
**13** This triangular prism shaped box needs a diagonal support inserted as between $A$ and $F$ as shown:
$AB = 19$, $BD = 30$ and $DF = 43$. find the length of $AF$ to two decimal places.




**14** From a point $15$ m due north of a tower, the angle of elevation of the tower is $32^\circ$.

**a** Find the height of the tower $h$. Round your answer to two decimals places.

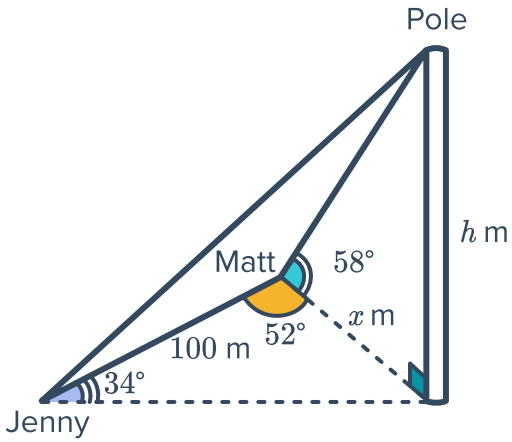
**b** Find the size $\theta$ of the angle of elevation of the tower at a point $20$ m due east of the tower. Round your answer to the nearest degree.




**15** A pole is seen by two people, Jenny and Matt:

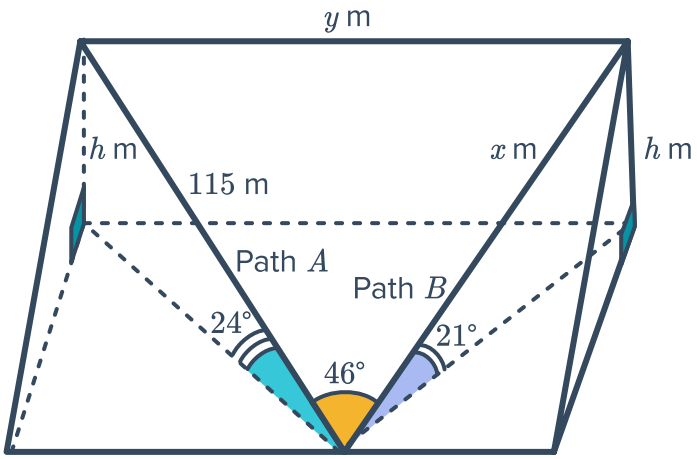
**a** Matt is $x$ m from the foot of the pole. Find $x$ to the nearest metre.

**b** Find the height of the pole $h$ to the nearest metre.




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**16** Two straight paths to the top of a cliff are inclined at angles of $24^\circ$ and $21^\circ$ to the horizontal:




**a** If path $A$ is $115\ \text{m}$ long, find the height $h$ of the cliff, rounded to the nearest metre.

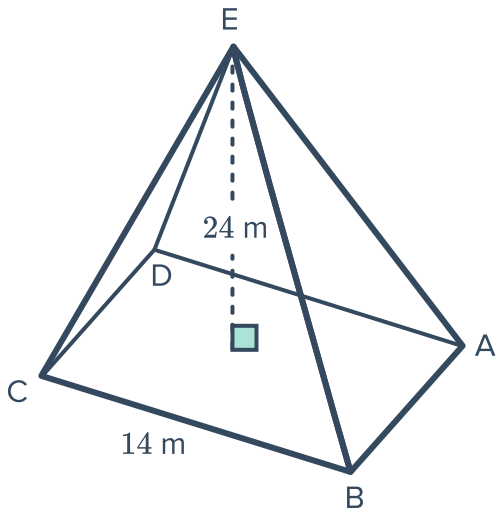
**b** Find the length $x$ of path $B$, correct to the nearest metre.

**c** Let the paths meet at $46^\circ$ at the base of the cliff. Find their distance apart, $y$, at the top of the cliff, to the nearest metre.

**17** A pyramid has a square base with side length $14\ \text{m}$ and a vertical height of $24\ \text{m}$:

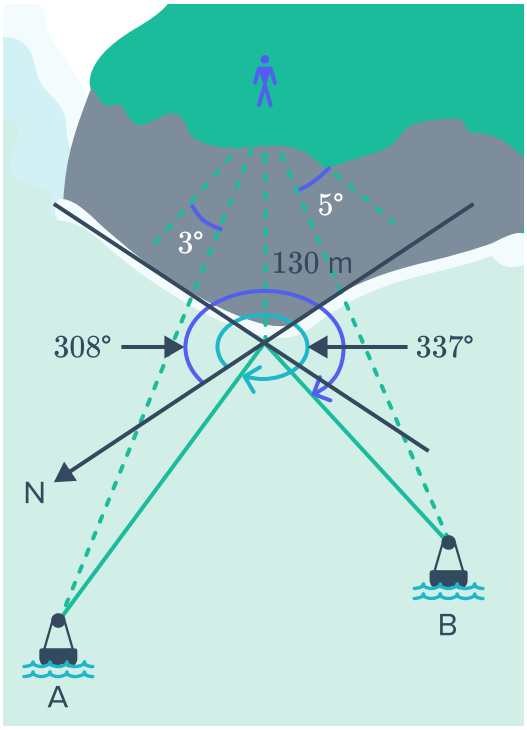
**a** Find the length of the edge $AE$. Round your answer to one decimal place.

**b** Find the size of $\angle BEA$, to the nearest minute.




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**18** Two buoys, $A$ and $B$, are observed from a lookout at the top of a $130\text{ m}$ high cliff. The bearing of buoy $A$ is $337^\circ$ and its angle of depression is $3^\circ$. The bearing of buoy $B$ is $308^\circ$ and its angle of depression is $5^\circ$:

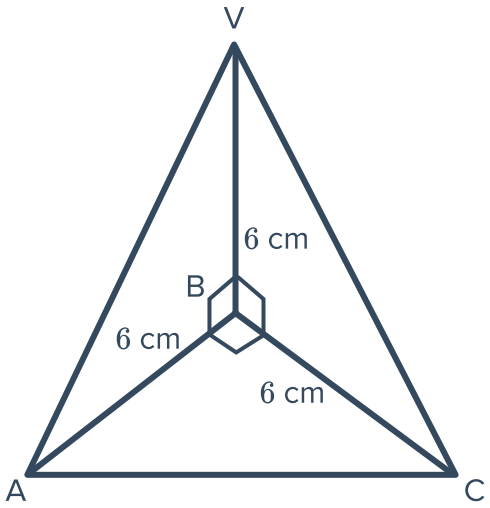



**a** Find the distance, to the nearest metre, from buoy $A$ to the base of the cliff.

**b** Find the distance from buoy $B$ to the base of the cliff, to the nearest metre.

**c** Find the distance between the two buoys, to the nearest metre.

**19** In the tetrahedron shown, the angles $\angle VBC$, $\angle VBA$, and $\angle ABC$ are all right-angles.

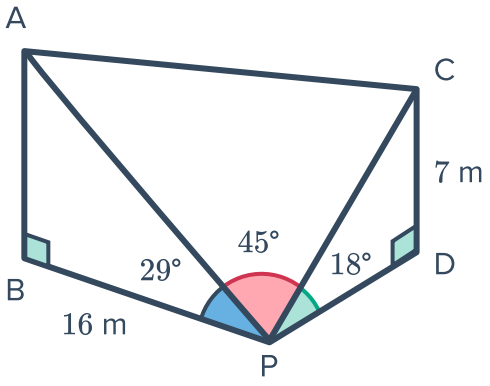



**a** Find the following distances, to two decimal places:

i $VA$ ii $VC$ iii $AC$

**b** Find the size of angle $\angle VCA$. Round your answer to the nearest minute.

**20** Roald is standing at point $P$ and observes two poles, $AB$ and $CD$, of different heights. $P$, $B$, and $D$ are on horizontal ground: From $P$, the angles of elevation to the top of the poles at $A$ and $C$ are $29^\circ$ and $18^\circ$ respectively. Roald is $16\text{ m}$ from the base of pole $AB$. The height of pole $CD$ is $7\text{ m}$.




**a** Calculate the distance from Roald to the top of pole $CD$, to two decimal places.

**b** Calculate the distance from Roald to the top of pole $AB$, to two decimal places.

**c** Calculate the distance between the tops of the poles, to two decimal places.

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**21** Point $A$ is due north of a tower, and has an angle of elevation to the top of the tower of $51^\circ$. Point $B$ is $100\ \text{m}$ from point $A$ on a bearing of $120^\circ$. The angle of elevation from point $B$ to the top of the tower is $25^\circ$.
Find the height of the tower to the nearest metre.

**22** A room measures $5\ \text{m}$ in length and $4\ \text{m}$ in width. The angle of elevation from the bottom left corner to the top right corner of the room is $57^\circ$.

**a** Find the exact distance from one corner of the floor to the opposite corner of the floor.

**b** Find the height of the room. Round your answer to two decimal places.

**c** Find the angle of elevation from the bottom corner of the $5\ \text{m}$ long wall to the opposite top corner of the wall. Round your answer to two decimal places.

**d** Find the angle of depression from the top corner of the $4\ \text{m}$ long wall to the opposite bottom corner of the wall. Round your answer to two decimal places.

**23** $A$ and $B$ are two positions on level ground. From an advertising balloon at a vertical height of $740\ \text{m}$, $A$ is observed in an easterly direction and $B$ at a bearing of $159^\circ$. The angles of depression of $A$ and $B$, as viewed from the balloon, are $30^\circ$ and $20^\circ$, respectively.

**a** Find the distance from point $A$ to the point directly below the advertising balloon, to the nearest metre.

**b** Find the distance from point $B$ to the point directly below the advertising balloon, to the nearest metre.

**c** Find the distance between points $A$ and $B$, to the nearest metre.

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